Morning Star Project · Theorema Aureum 143

Certificate of Unconditional Formal Proof

h(Q(√-143)) = 10 and ClassGroup(OK) = <[p2]>

Issued 2026-06-23 · 0 sorry · 0 admitted · axiom footprint: classical trio

§1 Theorem Statements

Let **K = Q(sqrt(-143))** and OK its ring of integers. Two theorems are proved unconditionally in Lean 4 (Mathlib v4.12.0), 0 sorry, axiom footprint = classical trio:

Theorem A — class number:

```
classNumber K = Fintype.card (ClassGroup (OK)) = 10
```

Theorem B — cyclic generator (proved 2026-06-23):

```
forall x : ClassGroup (OK), x in Subgroup.zpowers [p2]
```

Together: **ClassGroup(OK) is cyclic of order 10, generated by [p2]**, the prime ideal above 2 with Ideal.absNorm p2 = 2.

§2 Proof Architecture

Pillar	Lean theorem	Key step
Lower bound	BSD_classNumber_lower_bound	orderOf [p2] = 10 >= classNumber K (Lagrange)
p2^10 principal	BSD_p2_pow_10_principal	Norm-form cert: (2a+b)^2+143b^2=4*2^10, Z-basis
classNumber = 10	BSD_classNumber_eq_10_via_principal	Lagrange-divisibility bypass; no BQF API needed
Generator STAR NEW	BSD_classGroup_gen_by_p2_CLOSED	orderOf=10=|G| => zpowers [p2] = top (eq_top_of_card_eq)

§3 Generator Proof — BSD_classGroup_gen_by_p2_CLOSED

File: BSD_ClassGroup_Generator_CLOSED.lean | 0 sorry | classical trio

Step 1: hord : orderOf g = 10

via BSD_orderOf_p2_eq_10 BSD_p2_pow_10_principal

Step 2: hcn : classNumber K = 10

via BSD_classNumber_eq_10_via_principal BSD_p2_pow_10_principal

Step 3: orderOf g = Nat.card (ClassGroup OK)

via hord + hcn + Nat.card_eq_fintype_card

Step 4: Nat.card (zpowers g) = Nat.card G

via Nat.card_zpowers [Mathlib ZMod/Quotient.lean]

Step 5: zpowers g = top, then forall x, x in zpowers g

via Subgroup.eq_top_of_card_eq [Finite H] + Subgroup.mem_top

Proof term (abbreviated):

```
theorem BSD_classGroup_gen_by_p2_CLOSED : BSD_classGroup_gen_by_p2_hyp := by
have hord := BSD_orderOf_p2_eq_10 BSD_p2_pow_10_principal
have hcn  := BSD_classNumber_eq_10_via_principal BSD_p2_pow_10_principal
have hcard : Nat.card (zpowers g) = Nat.card (ClassGroup OK) :=
Nat.card_zpowers.trans (by rw [hord, Nat.card_eq_fintype_card]; exact hcn.symm)
have htop : Subgroup.zpowers g = top := Subgroup.eq_top_of_card_eq hcard
intro x; exact htop ■ Subgroup.mem_top x
```

§4 Surface Closure History

BSD_classNumber_lower_bound	**PROVED**	orderOf [p2] >= 10 >= classNumber K
BSD_p2_pow_10_principal	**PROVED**	p2^10 is principal (norm-form cert)
BSD_BQF_ClassNumber_bridge	**PROVED**	BQF bijection bypassed 2026-06-22
BSD_classNumber_eq_10_via_principal	**PROVED**	h(K) = 10, unconditional 2026-06-22
BSD_classGroup_gen_by_p2_hyp STAR	**PROVED**	Generator theorem 2026-06-23

§5 Axiom Footprint

Every theorem uses exactly the classical trio — no research-grade axioms:

- `propext`
- `Classical.choice`
- `Quot.sound`

Verified by #print axioms BSD_classGroup_gen_by_p2_CLOSED.

§6 SHA-256 File Manifest

File	SHA-256 (first 40 hex)
`BSD_ClassGroup_Generator_CLOSED.lean  STAR NEW`	`7a2b3699e7638c1fe3e788762dd68548081e889a`
`BSD_BQF_Bridge_Closed.lean`	`c39188955a73d388757b091563fb432ee69120a3`
`BSD_ClassNum_Upper_CLOSED.lean`	`a148717cf1a0ae49c8fda7309a4863600a1425b5`
`BSD_P2_Principal_CLOSED.lean`	`d2cbf0b7f7f57ad1c214095060e3b6a17119d27d`
`BSD_ClassNumberLowerProof.lean`	`42324524ee93a721073ceffa2a99c5d9e5927e8b`
`BSD_ReducedForms.lean`	`a782edecf6df66371c7a29449f48d98df13a093a`

§7 Repository Links

- **BSD proof tower:** https://github.com/DavidFox998/Birch-and-Swinnerton-Dyer
- **ClassNumber-143:** https://github.com/DavidFox998/ClassNumber-143
- **Certifications:** https://github.com/DavidFox998/Certifications

§8 Honest Scope

This certificate covers only h(Q(sqrt(-143))) = 10 and the generator [p2]. The Birch-Swinnerton-Dyer conjecture, Riemann Hypothesis, Yang-Mills mass gap, and Navier-Stokes regularity remain OPEN. No Clay Prize claim is made.